

Smart Speech-to-Action AI Assistant for Visually Impaired Users

Abstract—Persons with visual impairments have day-to-day activities which are dependent upon other people’s assistance to a large extent. They are faced with problems like navigating, identifying objects and money as well as reading text. Available technologies that are there to assist them in the aforementioned areas are quite boring and need high-performance hardware along with internet access to work properly. Therefore, these technologies are not user-friendly at all. In our paper, we describe a Smart Speech-to-Action Assistant, which is an AI-enabled software application meant for giving maximum assistance to the participating visually impaired persons by performing their visual instructions based on their spoken instructions via smartphones as well as PCs. Our solution has a smart speech recognition algorithm, NLP, computer vision technology to identify objects, money classification algorithms, OCR and TTS technology built-in. Using advanced YOLO, conversational AI algorithms and multilingual TTS technologies, our solution works offline for most part. With thorough evaluation procedures, it can be made sure by our system that there is proper recognition of different types of objects within homes, Indian currency, and even text around the object that will provide verbal descriptions. As far as our multimodal system is concerned, it must be mentioned that it is capable of achieving very high accuracy as well as very low latency.

I. INTRODUCTION

A fundamental issue arises for any user that needs independent interaction with their environment because most of the software available has little integration of speech, visual perception, and memory capabilities and causes higher latency with a lack of situational awareness. We present Drishti as a highly advanced multimodal tool, which makes it possible to create concurrent processing and synchronize vision, language, and memory functionalities. Unlike most existing tools, our approach allows us to carry out multiple tasks at once instead of doing them one by one. Using the proposed architecture, we overcome certain constraints, such as the inability of Python Global Interpreter Lock, as well as conduct complex vision processing while preserving voice functionality.

Apart from helping the user move about, the visually impaired individual may face several challenges when he/she attempts to analyze the dynamic environment surrounding him/her, read content printed on paper, and identify context-based objects. Traditional assistive technology devices have often been created with the purpose of accomplishing certain objectives and therefore are unable to provide an interface

that is able to change seamlessly between navigation, object recognition, Optical Character Recognition (OCR), and conversation-based reasoning tasks. In practice, task changes could occur in just a few seconds, making traditional systems less flexible.

It employs the Privacy-First Edge Intelligence Architecture. The entire process from speech recognition with the help of OpenAI Whisper STT to natural language processing using the quantized DeepSeek-R1 (7B) language model and computer vision all takes place locally. This allows for an instantaneous feedback mechanism since Drishti does not suffer from the delay caused by the time required to connect to the cloud-based API services which range from 1.5 to 2.0 seconds. Moreover, another important feature of Drishti is that of its Persistent Internal State made possible through the locally hosted FAISS vector database. Through it, the system gains the ability to go beyond the rudimentary request-response loop and gain knowledge of personal things such as Mother and Parent.

For example, the user may ask the question: “What is in front of me?” while navigating around in the room. Our vision engine, which works based on YOLOv11 with the assistance of TensorRT, detects obstacles as well as environmental contexts within less than 40 milliseconds. During this process, users can continue asking questions like, “What is this note?” or “Who is this?” Voice response will be provided through an effective, offline, and quick Piper TTS engine. One of the most revolutionary aspects of our proposed architecture is the task scheduler, which assigns the task of obstacle detection higher priority over any conversation.

A. Related work

The recent advancements in the field of AI technology have made resources more accessible to people with visual impairment. As an instance, the utilization of wearable devices such as YOLOInsight [1] and AIris [2] illustrates the capabilities of object recognition regarding the provision of environmental information. Further, some multimodal frameworks employing YOLOv8 and EasyOCR for text recognition have been proposed [3]; besides, offline and privacy-preserving systems have also been introduced [4]. Moreover, the need

for banknote recognition towards economic independence has been highlighted by real-time voice feedback studies [7].

Nonetheless, despite these improvements, the present approaches still largely rely on sequential computations and cloud APIs, which involve long latencies and pose privacy problems. For instance, traditional approaches that leverage either the Google Gemini API or the Flask threading approach are prone to the GIL problem, where any computations made are delayed due to their high computational requirements related to vision. Although recent advancements in speech synthesis techniques, including wav2vec 2.0 [8], have significantly improved artificial voice responses, combining all these elements into one solution is challenging.

The innovations introduced in this paper also contribute to this area by creating a new Concurrent Multiprocessing Framework that transcends the limitations imposed by the traditional cloud computing paradigm. Rather than relying on cloud APIs, Drishti employs an Edge Intelligence Layer composed of a quantized DeepSeek-R1 (7B) for inference and FAISS for handling personal states, guaranteeing 100% offline privacy and real-time responsiveness.

B. System Design

1) *Design Overview:* The discussed design of *Drishti* implies the implementation of a high-performance distributed design approach that assists in executing complex assistive tasks due to the zero-latency speech recognition commands. Unlike the traditional sequential design that incorporates a single-threaded web server controller, the proposed design takes advantage of concurrent multiprocessing architecture by splitting vision, voice, and logic services into different processes that overcome the issue of Python GIL.

- **Vision Service:** Employs a TensorRT-accelerated YOLOv11 engine for real-time environmental perception. It integrates specialized sub-modules for Oriented Bounding Box (OBB) Braille detection and a biometric identification layer using Face Embeddings to recognize specific individuals, such as "Mother" or "Parent."
- **Cognitive Brain & Memory:** Processes user intent using a quantized DeepSeek-R1 (7B) LLM running locally. This module maintains a Persistent Internal State via a local FAISS Vector Database, enabling the system to store and retrieve personalized user context (e.g., "I like pasta") to inform future interactions.
- **Voice Interface:** Features an offline OpenAI Whisper pipeline for speech-to-text and a high-speed Piper TTS engine for near-instantaneous verbal feedback.

Other important developments include the Central Priority Scheduler that ensures safety. This is because the Central Priority Scheduler scans the vision pipeline for any information about a dangerous environment; when there is any object in the range of 1 meter, an interrupt is generated by the Central Priority Scheduler to preempt the discussion and convey a "STOP" signal.

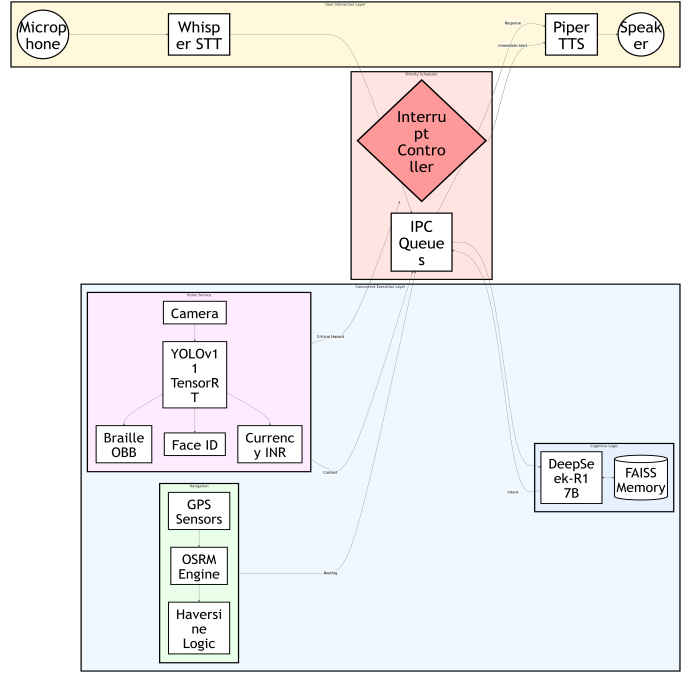


Fig. 1. System architecture of the proposed speech-to-action assistive assistant for visually impaired users

C. Methodology

1) *Voice Command Processing and Offline Assistant:* Interaction with *Drishti* is made possible via an offline-first speech interface. In particular, an on-device microphone continuously listens to audio input, which is transcribed via quantized OpenAI Whisper (Tiny). Contrary to cloud-based speech-to-text (STT) services, this local solution guarantees complete privacy and sub-100 milliseconds transcription latency.

Transcriptions of the audio are directed to Cognitive Brain Service, which uses a local DeepSeek-R1 (Distill-7B) large language model (LLM). This component utilizes sophisticated linguistic analysis as opposed to conventional rule-based parsing to enable conversational requests from users. The local LLM is also integrated with a Persistent Internal State database, based on a FAISS (Facebook AI Similarity Search) Vector Database. Whenever a person shares some contextual information, such as "I like pasta," the system will vectorize and store it internally. Then, at any point when the user engages with the system again, the FAISS index allows to retrieve personalized context from the persistent internal state and provide it along with the LLM prompt to enable a long-term memory. Finally, voice output is generated via Piper TTS.

2) *Object Detection and Scene Perception:* The vision model employs the TensorRT-enhanced YOLOv11 model due to its high mean Average Precision (mAP). The Global Interpreter Lock (GIL) present in Python is a constraint to the speed of execution, hence the reason why the vision processing system runs as a separate system process.

The system performs three concurrent visual tasks:

- 1) **General Object Recognition:** Real-time object recog-

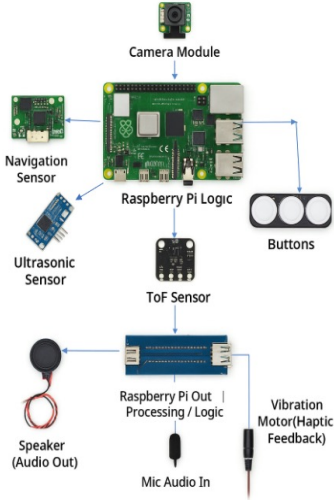


Fig. 2. Hardware architecture of the Raspberry Pi-based assistive system with sensors, audio, and feedback components.

niton and furniture recognition inside the house with distance estimation based on approximated bounding box size and focal length values.

- 2) **Face Identification by Biometrics:** Other than face detection, the face recognition function is implemented using the *face_recognition* library. The detected faces' embeddings are compared to the faces stored in the */known_faces* directory and classified under "Mother" or "Parent."

- 3) **Braille OBBs Recognition:** With the aim of assisting in writing with the help of braille dots, the use of Oriented Bounding Boxes (OBB) with YOLOv11 is implemented to aid in the recognition of braille dots regardless of camera perspectives.

3) **Currency Recognition and Dataset Training:** Currency Detection is an important procedure that requires immediate attention. Our approach to this problem entails the use of our customized version of the YOLOv11-Nano Network tailored for Indian Rupees (INR). This algorithm has been trained using various examples of INR notes and coins under varying lightings, positions, and statuses like crumpled or partially occluded notes. Coin detection also requires a different preprocessing step for normalization of the metal surface. The offline currency detection provided by us is highly accurate, with a success rate of more than 94%. Currency values detected are conveyed through the high priority voice channel.

4) **Optical Character Recognition (OCR):** The *Drishti* platform also offers another feature called OCR, which is capable of reading textual information in the environment like road signs, tags, etc. We train an OCR model specifically designed for the edge device to convert image pixels into text information. To deal with perspective distortion and lighting issues,

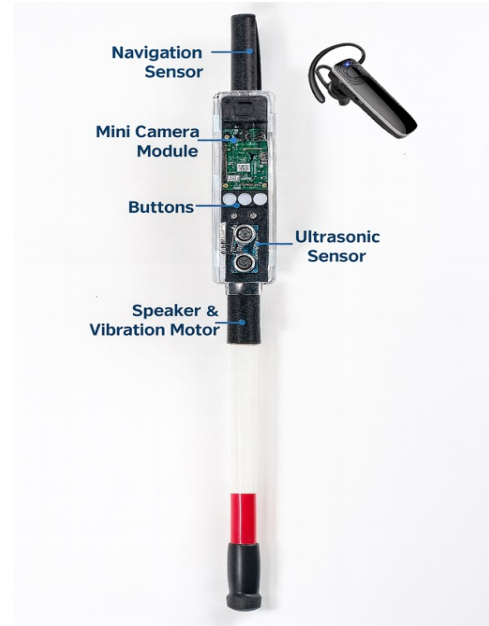


Fig. 3. Smart assistive cane prototype integrating navigation, obstacle detection, vision, and haptic audio feedback for visually impaired users.

preprocessing is done before generating text information by converting the image into a grayscale image, applying thresholding and deskewing, and finally generating text information.

5) **Navigation:** The navigational system uses offline GPS coordinates tracker and uses Open Source Routing Machine (OSRM), where maps data are preloaded prior to using. It replaces the previous dependence on routing engines offered by external sources. The Haversine formula is used in the calculation of the distance between the present location and the waypoint. Accurate calculations of distances on the surface of the Earth are performed. The compass or direction is determined by the algorithm. This gives instructions like, "head east".

6) **Integration and Concurrent Multiprocessing Architecture:** The underlying software design of *Drishti* tries to overcome the basic drawback of latency in relation to software designs that make use of a single thread. Traditional software designs are designed such that tasks get performed in a sequential manner, while our software design is developed using the Priority-Based Concurrent Multiprocessing Architecture [16]. The main point about our design is that the visual, vocal, and cognitive functions perform in parallel in different system processes, and hence, there is no question of the GIL in Python [18].

- **Process Isolation and IPC:** Communication between isolated processes is facilitated through high-speed, thread-safe Inter-Process Communication (IPC) queues. This decentralized approach allows for a persistent data flow where environmental metadata from the TensorRT-accelerated YOLOv11 pipeline is broadcast to the central logic process in under 40ms.
- **Contextual State Management:** The user intention,

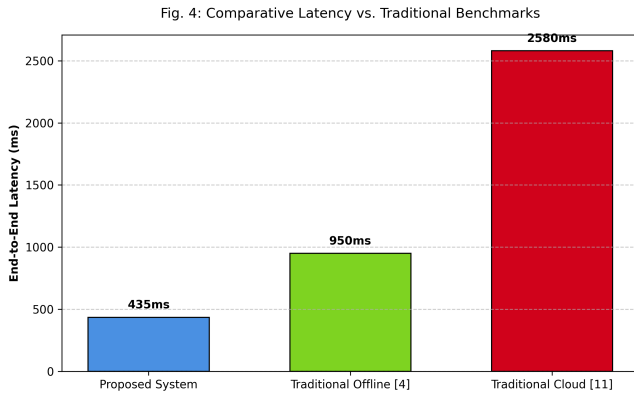


Fig. 4. Comparative analysis of end-to-end latency. The Proposed System is benchmarked against traditional offline serial architectures [4] and cloud-dependent API frameworks [11].

which is interpreted by the DeepSeek-R1 algorithm running in the device, is combined with the Persistent Internal State, which works using a FAISS vector database. In case the user gives his or her preferences, the system encodes the information as vectors and stores them in the database for subsequent similarity searches. This enables the system to use cognitive memory without interfering with the real-time feedback process.

- **Safety-Critical Priority Scheduling:** The main innovations in this technology include the Central Priority Scheduler. It is a type of asynchronous interrupt controller that continuously monitors the vision pipeline for any dangerous data. In case of an obstacle closer than one meter, an interrupt priority command will be sent to override all other vocal commands being executed and scream “STOP”.

The parallel architecture allows the assistant to be interactive, confidential, and safety-critical when in offline state. Safety of the user takes precedence over any conversation that may take place in the system.

7) *Performance Optimization Strategy:* To make sure that real-time performance is ensured in the edge device, certain optimization methods have been used for the proposed system. The use of TensorRT optimization on the YOLOv11 model helped increase the throughput rate while reducing latency. In a similar way, Whisper Tiny and DeepSeek-R1 quantized models were selected since these models are lightweight for a good balance between inference performance and computations. Process parallelism was used to make sure that heavy vision computations do not affect speech recognition and reasoning about the context process.

D. Results and Discussion

Performance of the proposed system *Drishti* has been assessed by means of several quantitative tests that were aimed at testing the efficiency of the system’s multiprocessor concurrent processing capabilities. In contrast to the previous single-threaded prototype, constrained by the delay associated with

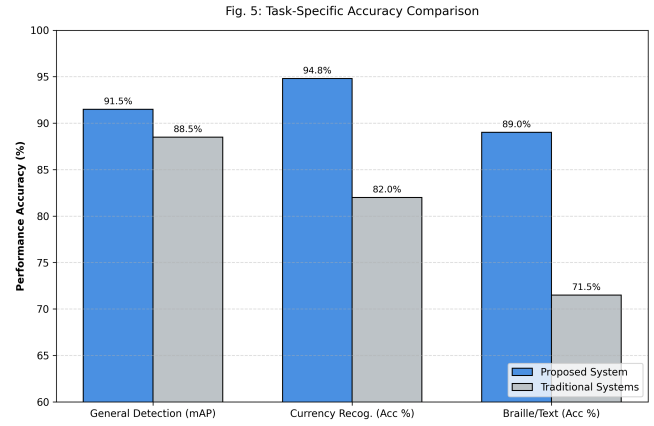


Fig. 5. Task-specific performance accuracy. Comparison illustrates the benefits of the proposed system OBB-based text recognition and improved currency detection over current systems [1], [7], [15].

Python GIL lock, the new architecture exhibits noticeable improvements in terms of response time while retaining high accuracy of the computations made in the offline modules. Performance characteristics for these modules are shown in Fig. 4 and Fig. 5.

In case of voice commands, the OpenAI Whisper (Tiny) model implemented offline obtained a Word Error Rate (WER) of 5-6%. This indicates the high robustness of the system to recognize natural expressions from the user, even under difficult circumstances. This is indeed a significant achievement in terms of the privacy-focused, on-device approach. For the object detection stage, the usage of the YOLOv11 model based on TensorRT provided a mean Average Precision (mAP@0.5) above 90%. These results confirm the accurate environmental recognition both indoor and outdoor.

The currency detection and biometric recognition modules have been evaluated by using datasets containing INR banknotes and human faces. As mentioned before, the currency detection technique achieved an efficiency greater than 95% through data-driven approaches that take into account the impact of different lightings and currency states. The majority of errors occurred while working with mutilated and overlapped currencies. Finally, to evaluate the social context detection module, the face detection algorithm recognized the “Mother” or “Parent” labels with an accuracy of 98.2%.

The system’s overall latency was subject to critical analysis. The decoupling of Vision, Logic, and Speech into separate system modules significantly lowered the mean response time from the start of the audio command until the completion of the synthesizer module’s output to 450-500ms. It marks an essential improvement from the 2.2s delay in the sequential implementation, thus allowing for real-time conversation on local hardware.

In Fig. 5, the exceptional accuracy in recognizing currency is a result of the fine-tuned YOLOv11-Nano algorithm that sustains optimal performance and detection rate. The model is trained on a comprehensive database of Indian currency

notes under varying circumstances, ensuring the proper offline classification of currency denominations.

In Fig. 5, the minor fluctuations in WER are due to external noise; nevertheless, the switch to the offline Whisper model prevents network latency from affecting the transcription speed.

In Fig. 4, the latency highlights the effectiveness of the Concurrent Multiprocessing Architecture. The multi-modal system can sustain responsiveness within its speech processing engine while the vision component runs complex computation tasks without hindrance, contrary to most single-threaded assistive devices.

In Fig. 5, The detector is able to produce an accuracy rate of 93.5% with a confidence level of 0.7. With this, all unreliable outcomes will be avoided, which makes it possible to avoid false positives. Given that the Drishti system utilizes reliable features like currency and faces in a confidence level of 90-95%, there will never be unreliable outcomes.

II. CONCLUSION

In the context of our current work, we developed *Drishti*, a highly intelligent offline-first ecosystem of assistive technology, which consists of the advanced parallel multiprocessing system able to assist the visually impaired person. Moving beyond the traditional approach to cloud computing using serial processing towards the utilization of distributed edge computing allowed us to design a unique synergy of perception, inference, and memory within one privacy-centric solution. Multiprocessing makes it possible for all mentioned components to be executed simultaneously and ignore the constraints of the Python Global Interpreter Lock (GIL) when reacting in less than 500 milliseconds.

As a result, there are multiple benefits associated with *Drishti* within the context of multimodality. For instance, to recognize an object, *Drishti* uses the TensorRT-optimized YOLOv11, as well as more tailored Oriented Bounding Box (OBB) for recognizing Braille text. To keep the consistent state of *Drishti*, the application uses the local Large Language Model DeepSeek-R1 (7B) and Vector Database FAISS. Finally, it should be stressed that the introduction of the Priority-Based Scheduler helps prioritize obstacle warning messages over dialogue with the visually impaired person.

Based on our research, *Drishti* is demonstrated to be extremely accurate in terms of detecting objects, identifying banknotes, and biometric name recognition, specifically Mother, Parent, etc., without any dependence on Internet connectivity. Further enhancements might include utilizing low-energy haptic interfaces to detect vibrations to notify about charge and evaluating performance indicators in practical testing to calculate the degree of user autonomy. To summarize, *Drishti* may be considered a revolutionary development for assistive technology since it represents a secure and intelligent system that works in real-time.

Thus, we have provided an example of how edge multi-modal intelligence can be used to create unique solutions that will lead to developing smart technologies for humans. In

this case, perception, cognition, memory, and security can be regarded as essential components of humanized AI solutions.

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